

AHAAN DAHIYA

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PROFESSIONAL SUMMARY

I am an Electrical Engineering undergraduate at Thapar Institute of Engineering and Technology who has independently built substantial hands on experience in Python programming, spanning deep learning, computer vision, and backend development. I enjoy writing clean and well documented code, I am comfortable working remotely with flexible hours, and I am looking for an internship where I can apply my existing Python skills to real world projects while learning new frameworks and best practices as part of a team.

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript
- Python Libraries and Frameworks: PyTorch, OpenCV, MediaPipe, scikit-learn, NumPy, pandas, BioPython, Streamlit
- Backend and Web Technologies: Node.js, Express.js, REST API design
- Machine Learning and Computer Vision: convolutional neural networks, K nearest neighbor classification, one hot encoding, real time video processing, rule based classification systems
- Tools and Practices: Git, GitHub, debugging, code review, technical documentation

PROJECTS

GeneLens-AI, a DNA Promoter Classifier

I built a deep learning pipeline in Python and PyTorch that determines whether a given DNA sequence is a promoter region using a convolutional neural network. The pipeline relies on BioPython to parse FASTA formatted genomic files and encodes each sequence with one hot vectors before feeding it into the trained model. I also built a Streamlit web interface so predictions can be tested interactively, and I organized the codebase into separate modules for training, evaluation, and inference.

GitHub: <https://github.com/ahaandahiya-cmd/GeneLens-AI>

Sign Language Translator, an ASL Fingerspelling Recognition App

I developed a real time computer vision application in Python that reads American Sign Language fingerspelling from a webcam feed and converts it directly into typed text. Using MediaPipe for hand landmark detection and OpenCV for video capture, I engineered geometric hand shape features and built both a rule based classifier that works immediately without training and an optional K nearest neighbor model that can be trained on a user's own hand samples using scikit-learn.

GitHub: https://github.com/ahaandahiya-cmd/Sign_Language_Translator

GeoTrade, a Geopolitical Market Signal Dashboard

I built a full stack web application in Node.js and Express that turns publicly available geopolitical news into localized market signal explanations. The backend ingests RSS and GDELT news feeds, deduplicates overlapping coverage, and classifies each story using a locally built keyword and entity lexicon rather than any external AI service. On the frontend, I built an interactive world map alongside a live quote watchlist and a transparent trade cost calculator.

GitHub: https://github.com/ahaandahiya-cmd/Global_Trade_News

EDUCATION

Bachelor of Engineering in Electrical Engineering, Thapar Institute of Engineering and Technology
First Year, Semester 1 | Expected graduation around 2030

CERTIFICATIONS

Python and Prompt Engineering Certification, Shivaji College, completed July 2026